

Valuation of Options - part 2

Quantitative Finance

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Assumptions Black-Scholes market (recap)

In class we will almost exclusively work with the 'Black-Scholes market'.

Assumptions on price processes:

Asset price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad \text{var}(W_1) = 1$$

Money Market Account:

$$dB_t = rB_t dt, \quad B_0 = 1$$

Assumptions on market:

- frictionless trading
 - no transaction costs
 - trading in continuous-time possible
 - no restrictions on short sales and fractional positions
- borrowing rate = lending rate

- we are interested in price C_t of (European) option with payoff $C_T = h(S_T)$
- we will discuss three methods to determine fair price (using continuous-time model for financial market):
 - Black-Scholes Partial Differential Equation
 - risk-neutral pricing
 - pricing kernel

These slides discuss the Black-Scholes Partial Differential Equation approach.

Section 1

The Black-Scholes Partial Differential Equation

The problem

Setup:

- Black-Scholes market (1 risky asset)
- we restrict to Markovian trading strategies:

$$\phi_t = \tilde{\phi}(B_t, S_t) = \phi(t, S_t) \text{ and } \psi_t = \tilde{\psi}(B_t, S_t) = \psi(t, S_t)$$

So we can write, for some function F ,

$$V_t = \phi(t, S_t)S_t + \psi(t, S_t)B_t = F(t, S_t)$$

Question:

Which F 's correspond to **self-financing** trading strategies?

Why interesting?

- consider option with payoff $h(S_T)$ at maturity
- if we can find self-financing portfolio with $F(T, S_T) = h(S_T)$ then no-arbitrage price of option at time $0 \leq t < T$ is given by

$$C_t = F(t, S_t)$$

Black-Scholes Partial Differential Equation

Black-Scholes Partial Differential Equation:

F corresponds to a self-financing trading strategy if and only if F is a solution to the Partial Differential Equation (equation with function G as variable)

$$\frac{\partial G}{\partial t}(t, s) + rs \frac{\partial G}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 G}{\partial s^2}(t, s) - rG(t, s) = 0 \quad \forall s > 0, t \in [0, T),$$

and in that case the positions in the self-financing trading strategy are given by:

$$\phi(t, s) = \frac{\partial F}{\partial s}(t, s),$$

and

$$\psi(t, s) = \frac{F(t, s) - \phi(t, s)s}{\exp(rt)}.$$

Remarks:

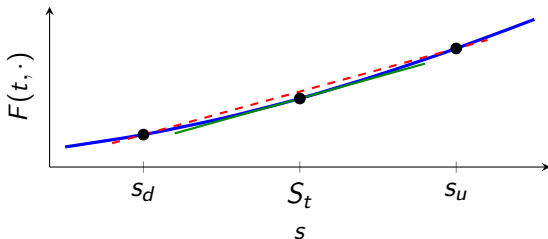
- μ does not play a role!
- using different models (i.e. SDEs) for B and S leads, in general, to different PDE!

From the binomial tree to the delta

In the one-period binomial model (part 1) the replicating stock position was a *difference quotient*,

$$\phi = \frac{c_u - c_d}{s_u - s_d} = \frac{F(t + \Delta t, s_u) - F(t + \Delta t, s_d)}{s_u - s_d},$$

the slope of the value across the two stock outcomes. In the PDE it is $\phi = F_s = \partial F / \partial s$, the **delta**: as the step shrinks, the secant slope $\rightarrow$ the tangent slope.



secant slope = $\frac{c_u - c_d}{s_u - s_d}$ (binomial) $\longrightarrow$ **tangent** slope = $F_s = \Delta$ (PDE).

Derivation of the PDE

Which F are possible (when using self-financing strategies $(\phi(t, S_t), \psi(t, S_t))$)? We have

(1) $V_t = F(t, S_t)$

(2) $dV_t = \phi(t, S_t) dS_t + \psi(t, S_t) dB_t$

- apply Itô to (1) and insert $dS_t \implies dV_t = a_t dt + b_t dW_t$
- insert dS_t and dB_t in RHS of (2) $\implies dV_t = c_t dt + d_t dW_t$
- obtain system of equations

$$\begin{cases} a_t = c_t \\ b_t = d_t \end{cases}$$

- solving $\implies$ conditions on F (PDE)

Warning: sloppy notation

We will often use the following notations/abbreviations when working with PDEs:

- $F_s = (\partial F / \partial s) = (\partial F / \partial s)(t, S_t)$
- $F_t = (\partial F / \partial t) = (\partial F / \partial t)(t, S_t)$
- $F = F(t, S_t)$
- etc.

The notation F_t is somewhat confusing as it seems to refer to a stochastic process. And you argue that it would be a natural abbreviation for $F(t, S_t)$.

If you want to be clear and safe it could be better to just write down the partial derivatives fully.

Derivation

We have

(1) $V_t = F(t, S_t)$

(2) $dV_t = \phi(t, S_t) dS_t + \psi(t, S_t) dB_t$

- apply Itô to (1) and insert $dS_t = \mu S_t dt + \sigma S_t dW_t \implies$

$$\begin{aligned} dV_t &= F_s dS_t + F_t dt + \frac{1}{2} F_{ss} d[S, S]_t \\ &= \left(F_s \mu S_t + F_t + \frac{1}{2} F_{ss} \sigma^2 S_t^2 \right) dt + F_s \sigma S_t dW_t \end{aligned}$$

- insert dS_t and $dB_t = rB_t dt$ in RHS of (2) $\implies$

$$dV_t = (\phi_t \mu S_t + r \psi_t B_t) dt + \phi_t \sigma S_t dW_t$$

- hence $\phi_t = \phi(t, S_t) = F_s(t, S_t) = F_s$
- and

$$F_s \mu S_t + F_t + \frac{1}{2} \sigma^2 F_{ss} S_t^2 = \phi_t \mu S_t + r \psi_t B_t$$

We need to prove

$$\frac{\partial F}{\partial t}(t, s) + rs \frac{\partial F}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 F}{\partial s^2}(t, s) - rF(t, s) = 0 \quad \forall s > 0, t \in [0, T)$$

On the previous slide we obtained

$$F_t + \frac{1}{2} \sigma^2 F_{ss} S_t^2 = r \psi_t B_t.$$

As $\psi_t B_t = F - \phi_t S_t = F - F_s S_t$ we obtain the result.

Alternative Derivation

- sell (write) and hold 1 option with price process $C_t = F(t, S_t)$ (assumption!)
- let ϕ_t be number of stocks at time t and ψ_t the position in the MMA
- yields portfolio value $V_t = -C_t + \phi_t S_t + \psi_t B_t$
- this portfolio is self-financing if

$$dV_t = -dC_t + \phi_t dS_t + \psi_t dB_t$$

- using Itô:

$$dV_t = -F_s dS_t - F_t dt - \frac{1}{2} F_{ss} d[S, S]_t + \phi_t dS_t + \psi_t r B_t dt$$

- can only eliminate local risk (dW_t) for $\phi_t = F_s(t, S_t)$!
- yields $dV_t = \dots dt$ which has (locally) no risk, so rate of return is same as on $B \implies dV_t = rV_t dt$
- hence

$$-F_t - \frac{1}{2} F_{ss} \sigma^2 S_t^2 + r\psi_t B_t = r(-F + F_s S_t + \psi_t B_t)$$

which yields Black-Scholes PDE

Application

Given is a European option with payoff $h(S_T)$ at expiration date/maturity T . How can we use the PDE to obtain the price of this option?

- Solve the PDE,

$$\frac{\partial G}{\partial t}(t, s) + rs \frac{\partial G}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 G}{\partial s^2}(t, s) - rG(t, s) = 0,$$

for all $s > 0$, $t \in [0, T)$, under the boundary condition

$$G(T, s) = h(s) \text{ for all } s > 0.$$

- If F is solution to PDE satisfying the boundary condition, then no-arbitrage implies that the price of the option at time $t \in [0, T)$ is given by $F(t, S_t)$.
- If you are lucky: 'closed-form' solution can be found
 - there is relation to well studied heat equation
- unlucky: use numerical techniques; see notebook

Examples: closed-form solutions

- $h(S_T) = S_T$

$$F(t, s) = s$$

- European call option: $h(S_T) = \max\{S_T - K, 0\}$

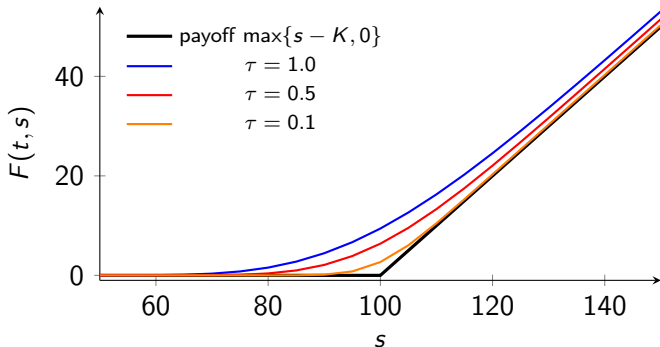
$$F(t, s) = s\Phi(d_1) - \exp(-r(T - t))K\Phi(d_2),$$

with $d_1 = d_2 + \sigma\sqrt{T - t}$ and

$$d_2 = \frac{\log(s/K) + (r - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$

The call price as a function of s

The PDE solution $F(t, s)$ for a European call ($K = 100$, $r = 3\%$, $\sigma = 20\%$), for several times-to-maturity $\tau = T - t$:



As $\tau \rightarrow 0$ the price collapses onto the payoff; the gap above it is the option's *time value*.

Worked example: verifying a candidate solution

To check that a function F solves the B-S PDE, just substitute it. Two quick cases (recall $F_t + rsF_s + \frac{1}{2}\sigma^2 s^2 F_{ss} - rF = 0$):

(a) the stock, $F(t, s) = s$: then $F_t = 0$, $F_s = 1$, $F_{ss} = 0$, so

$$0 + rs \cdot 1 + 0 - r s = 0. \quad \checkmark$$

(b) a zero-coupon bond paying K at T , $F(t, s) = K e^{-r(T-t)}$: then $F_s = F_{ss} = 0$ and $F_t = rK e^{-r(T-t)} = rF$, so

$$rF + 0 + 0 - rF = 0. \quad \checkmark$$

By *linearity* of the PDE, any combination $as + bK e^{-r(T-t)}$ also solves it — consistent with such a stock+bond portfolio being self-financing.

(In the exercises you verify that the digital-call solution $F(t, s) = e^{-r(T-t)} \Phi(d_2)$ solves the PDE.)

Quiz

A portfolio has value process $V_t = F(t, S_t)$ with

$$F(t, s) = 2s + 3K e^{-r(T-t)}.$$

- 1 Does F solve the Black-Scholes PDE?
- 2 Which self-financing strategy (ϕ, ψ) does it correspond to?
- 3 Which claim does the portfolio pay at maturity T ?

Examples: European digital call option

- payoff: $h(S_T) = 1\{S_T \geq K\}$
- the B-S PDE can be solved explicitly:

$$F(t, s) = \exp(-r(T - t))\Phi(d_2),$$

with

$$d_2 = \frac{\log(s/K) + (r - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$

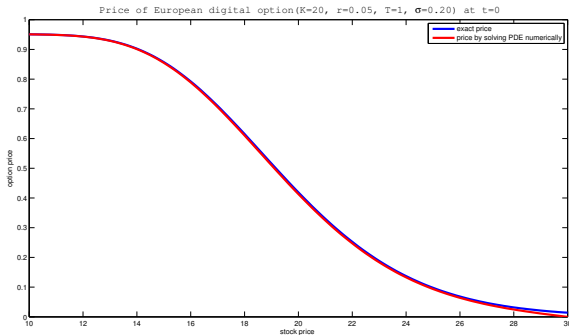
- Probability that option ends in-the-money:

$$\mathbb{P}\{S_T \geq K\} = \Phi\left(\frac{\log(s/K) + (\mu - 0.5\sigma^2)(T - t)}{\sigma\sqrt{T - t}}\right)$$

In the exercise set you will be asked to verify that F is a solution to the B-S PDE.

Examples: European digital put option

Price of the option (at $t = 0$) $F(0, s)$ - numerical solution compared to exact solution:

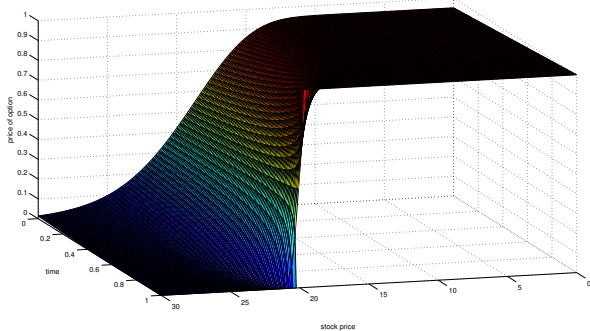


The Group Assignment might contain an exercise on solving the PDE numerically.

Examples: European digital put option

Value function $F(t, s)$:

Price of option as function of time and stock price European digital option, $K=20$, $T=1$, $r=0.05$, $\sigma=0.20$ (exact solution)



Solving the PDE numerically: finite differences

When there is no closed form, solve the PDE on a grid $t_0 < \dots < t_N = T$ (step Δt) and $s_0 < \dots < s_M$ (step Δs).
Replace the derivatives by differences:

$$F_t \approx \frac{F(t + \Delta t, s) - F(t, s)}{\Delta t}, \quad F_s \approx \frac{F(t, s + \Delta s) - F(t, s - \Delta s)}{2 \Delta s},$$
$$F_{ss} \approx \frac{F(t, s + \Delta s) - 2F(t, s) + F(t, s - \Delta s)}{\Delta s^2}.$$

- start from the known terminal values $F(T, s) = h(s)$ and step *backward* in time;
- the explicit scheme writes each $F(t, s_i)$ as a (discounted) weighted average of its three neighbours $F(t + \Delta t, s_{i-1})$, $F(t + \Delta t, s_i)$, $F(t + \Delta t, s_{i+1})$ — i.e. a *trinomial tree*.

An implementation — including the explicit/trinomial scheme and its stability condition — is provided in the accompanying notebook.

The explicit finite-difference scheme *is* a trinomial tree; the binomial tree of part 1 is its two-neighbour cousin. For the scheme to make sense the three weights must be **nonnegative** — they play the role of risk-neutral probabilities.

- nonnegative weights require Δt small relative to Δs^2 (a *stability* / *CFL* condition); otherwise the scheme oscillates and the “probabilities” turn negative;
- the same constraint shows up in the tree: the log-price step must scale like $\sigma\sqrt{\Delta t}$ (so $u = e^{\sigma\sqrt{\Delta t}}$) to keep $q \in (0, 1)$.

One picture (an analogy, not a literal equivalence): the discrete weights must stay nonnegative to act as risk-neutral probabilities — mirroring how a valid risk-neutral measure rules out arbitrage. The PDE, the trees, and the grid are three views of the same object.

From the Black-Scholes PDE to the heat equation

A change of variables makes the connection precise. Write $\tau = T - t$ (time-to-maturity) and $x = \log s$, and set $F(t, s) = v(\tau, x)$. Using $s \partial_s = \partial_x$ and $s^2 \partial_{ss} = \partial_{xx} - \partial_x$, the PDE turns into a *constant-coefficient* equation with an *initial* condition $v(0, x) = h(e^x)$:

$$v_\tau = \frac{1}{2} \sigma^2 v_{xx} + \left(r - \frac{1}{2} \sigma^2\right) v_x - r v.$$

Rescaling time as $\tilde{\tau} = \frac{1}{2} \sigma^2 \tau$ and substituting $v = e^{ax+b\tilde{\tau}} u$ with

$$a = -\frac{1}{2}(k-1), \quad b = -\frac{1}{4}(k+1)^2, \quad k = \frac{2r}{\sigma^2},$$

removes the first- and zeroth-order terms, leaving the pure **heat equation**

$$u_{\tilde{\tau}} = u_{xx}.$$

So pricing a European option is, after a change of variables, an initial-value problem for the heat equation.

Food for thought

The B-S PDE comes with a *terminal* condition $F(T, s) = h(s)$, so it runs *backward* in time. With $\tau = T - t$, $x = \log s$ (and a discount/scaling factor) it becomes the *heat equation*

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

- The heat equation instantly *smooths* its initial data. What does this say about the smoothness of the price $F(t, \cdot)$ for $t < T$, even when the payoff h has a kink (call) or a jump (digital)?
- Why is “backward in time, from maturity” the natural direction for valuation?

Feynman-Kac Theorem

Let $\alpha(t, x)$, $\beta(t, x)$, and $k(t, x)$ be functions satisfying 'regularity conditions'. Consider the PDE

$$\frac{\partial G}{\partial t}(t, x) + \alpha(t, x) \frac{\partial G}{\partial x}(t, x) + \frac{1}{2} \beta^2(t, x) \frac{\partial^2 G}{\partial x^2}(t, x) - k(t, x) G(t, x) = 0,$$

for all x and $t \in [0, T)$ with boundary condition $G(T, x) = h(x)$ for all x .

Then the solution is given by:

$$G(t, x) = \mathbb{E} \left[\exp \left(- \int_t^T k(s, X_s) ds \right) h(X_T) \mid X_t = x \right],$$

where X is a stochastic process, on the time interval $[t, T]$, defined via the SDE

$$dX_u = \alpha(u, X_u) du + \beta(u, X_u) dW_u$$

and with starting value $X_t = x$, and where W is a standard Brownian motion.

Sketch of proof (Feynman-Kac)

Let F solve the PDE with $F(T, x) = h(x)$, and let X follow $dX_u = \alpha(u, X_u) du + \beta(u, X_u) dW_u$ on $[t, T]$ with $X_t = x$. Define the discounted process

$$M_u = \exp\left(-\int_t^u k(v, X_v) dv\right) F(u, X_u), \quad u \in [t, T].$$

With $D_u = \exp(-\int_t^u k dv)$, Itô's formula gives

$$\begin{aligned} dM_u &= D_u \left[-k F du + (F_t + \alpha F_x + \frac{1}{2} \beta^2 F_{xx}) du + \beta F_x dW_u \right] \\ &= D_u \underbrace{\left(F_t + \alpha F_x + \frac{1}{2} \beta^2 F_{xx} - kF \right)}_{=0 \text{ (the PDE)}} du + D_u \beta F_x dW_u. \end{aligned}$$

So M has *no drift*: it is a (local) martingale, hence — under integrability conditions — a martingale. As $M_t = F(t, x)$ and $M_T = \exp(-\int_t^T k dv) h(X_T)$,

$$F(t, x) = \mathbb{E}[M_T \mid X_t = x] = \mathbb{E}\left[\exp\left(-\int_t^T k(v, X_v) dv\right) h(X_T) \mid X_t = x\right].$$

Black-Scholes PDE reconsidered

The no-arbitrage price, at time $t \in [0, T)$, of a European option with payoff $h(S_T)$ at expiration date $T > 0$ is given by $F(t, S_t)$ where F satisfies

$$\frac{\partial F}{\partial t}(t, s) + rs \frac{\partial F}{\partial s}(t, s) + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 F}{\partial s^2}(t, s) - rF(t, s) = 0 \quad \forall s > 0, t \in [0, T)$$

Feynman-Kac tells us that we have

$$F(t, s) = \mathbb{E} \left[e^{-r(T-t)} h(S_T) \mid S_t = s \right],$$

where S is defined via the SDE

$$dS_u = rS_u du + \sigma S_u dW_u, \quad S_t = s,$$

where W is a standard Brownian motion.

Uniqueness and the maximum principle

Two structural facts make “the” price well-defined and well-behaved.

- **Uniqueness.** Under mild growth conditions the B-S PDE together with the terminal condition $F(T, s) = h(s)$ has a *unique* solution. Hence $F(t, S_t)$ is unambiguous — the analytic counterpart of market *completeness*.
- **Maximum principle.** For this (parabolic) PDE a solution attains its extreme values on the boundary of the domain — here the terminal time and the s -boundaries. Two consequences:
 - *monotonicity*: if $h_1 \leq h_2$ then $F_1 \leq F_2$ (prices respect the ordering of payoffs);
 - *no-arbitrage bounds*, e.g. $0 \leq F_{\text{call}}(t, s) \leq s$.

Both are also immediate from Feynman-Kac:

$F(t, s) = \mathbb{E}[e^{-r(T-t)} h(S_T) \mid S_t = s]$ is linear and monotone in the payoff h .